

The Ethics of AI-Assisted Code Migration

Regulated financial systems: accountability, transparency & human oversight

Delegating code authorship does not delegate moral responsibility.

The stakes: miscalculated compensation, compliance failures and erosion of public trust.

01 ACCOUNTABILITY GAP

Who answers when the code fails?

Responsibility is assigned without the time or understanding to exercise it.

PROPOSED SAFEGUARD

MANDATORY HUMAN REVIEW

Scope → Validate → Certify

Risk-based scoping, component review and a named engineer’s final sign-off.

02 TRANSPARENCY DEFICIT

Can we reconstruct the conversion?

Opaque changes obscure the connection between original and migrated logic.

PROPOSED SAFEGUARD

PROVENANCE DOCUMENTATION

Source → Model → Edits → Reviewer

Record versions, run configuration, human changes and approval identity.

03 AUTOMATION BIAS

Does compiling justify confidence?

Plausible output and shallow tests can hide financially significant errors.

PROPOSED SAFEGUARD

DIFFERENTIAL TESTING

Same inputs → Compare outputs

Run original and migrated systems; review rounding, dates, nulls and currency.

RESEARCH SPOTLIGHT

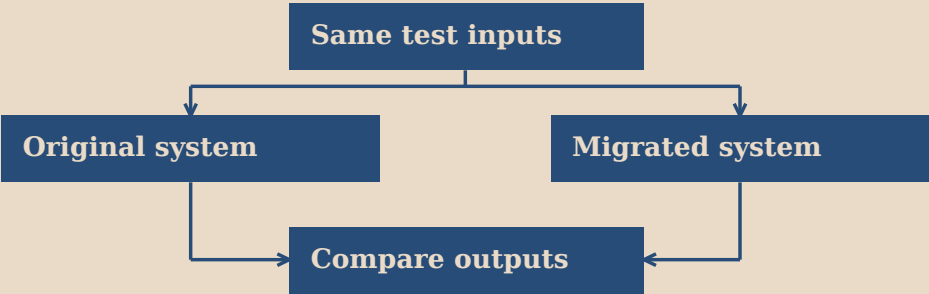
24 participants

Preference ≠ proven performance

Most preferred Copilot, yet understanding, editing and debugging remained difficult.

Vaithilingam et al. · CHI 2022

DIFFERENTIAL TESTING IN ACTION



Differences → investigate, correct & retest

RESEARCH SPOTLIGHT

Experts are susceptible too

Automation bias crosses experience levels

The review reports errors from both following incorrect advice and missing needed action.

Parasuraman & Manzey · Human Factors 2010